SPLUNK

Splunk is a powerful SIEM solution that provides the ability to search and explore machine data. **Search Processing Language (SPL)** is used to make the search more effective. It comprises various functions and commands used together to form complex yet effective search queries to get optimized results.

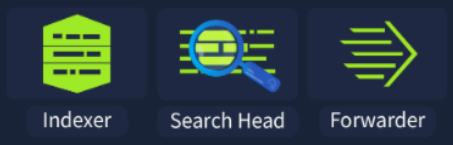
This room will dive deep into some key fundamentals of searching capability, like chaining SPL queries to construct simple to complex queries.

Learning Objectives

This room will teach the following topics:

What are Search processing Language?
How to apply filters to narrow down results.
Using transformational commands.
Changing the order of the results.

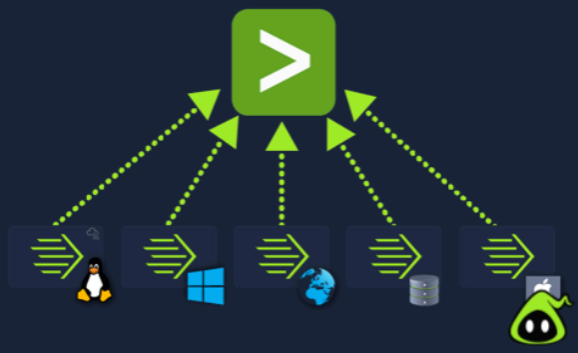
Splunk has three main components: Forwarder, Indexer, and Search Head. These components work together to help us search and analyze the data. These components are explained below:



Splunk Forwarder

Splunk Forwarder is a lightweight agent installed on the endpoint intended to be monitored, and its main task is to collect the data and send it to the Splunk instance. It does not affect the endpoint's performance as it takes a few resources to process. Some of the key data sources are:

- Web server generating web traffic.
- Windows machine generating Windows Event Logs, PowerShell, and Sysmon data.
- Linux host generating host-centric logs.
- Database generating DB connection requests, responses, and errors.



The forwarder collects the data from the log sources and sends it to the Splunk Indexer.

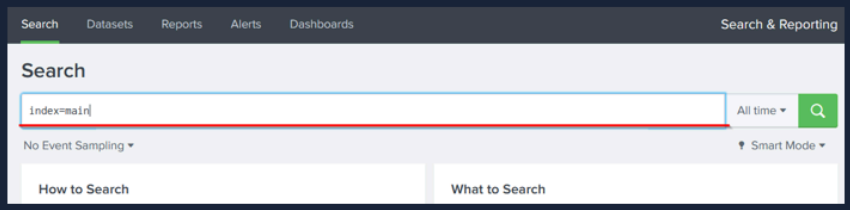
Splunk Indexer

Splunk Indexer plays the main role in processing the data it receives from forwarders. It parses and normalizes the data into field-value pairs, categorizes it, and stores the results as events, making the processed data easy to search and analyze.

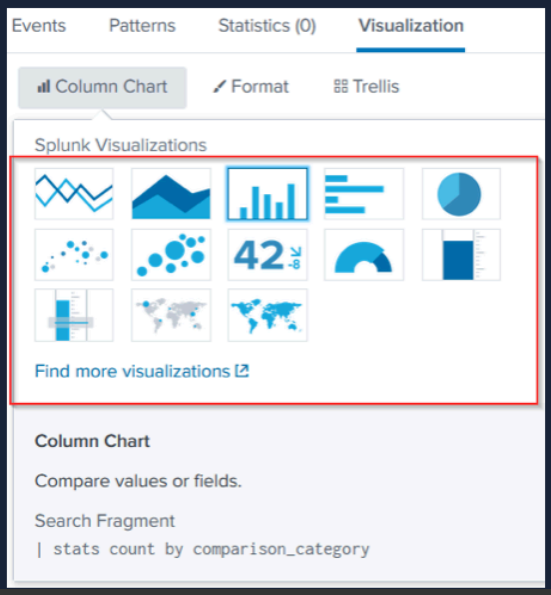
Now, the data, which is normalized and stored by the indexer, can be searched by the Search Head, as explained below.

Search Head

Splunk Search Head is the place within the **Search & Reporting App** where users can search the indexed logs, as shown below. The searches are done using the **SPL** (Search Processing Language), a powerful query language for searching indexed data. When the user performs a search, the request is sent to the indexer, and the relevant events are returned as field-value pairs.



The Search Head also allows you to transform results into presentable tables and visualizations such as pie, bar, and column charts, as shown below:



Splunk can ingest any data. According to the Splunk documentation, when data is added to Splunk, the data is processed and transformed into a series of individual events. The data sources can be event logs, website logs, firewall logs, etc. The data sources are grouped into categories.

Below is a chart listing from the Splunk documentation detailing each data source category.

Data source	Description
Files and directories	Most data that you might be interested in comes directly from files and directories.
Network events	The Splunk software can index remote data from any network port and SNMP events from remote devices.
IT Operations	Data from IT Ops, such as Nagios, NetApp, and Cisco.
Cloud services	Data from Cloud services, such as AWS and Kinesis.
Database services	Data from databases such as Oracle, MySQL, and Microsoft SQL Server.
Security services	Data from security services such as McAfee, Microsoft Active Directory, and Symantec Endpoint Protection.
Virtualization services	Data from virtualization services such as VMWare and XenApp.
Application servers	Data from application servers such as JMX & JMS, WebLogic, and WebSphere.
Windows sources	The Windows version of Splunk software accepts a wide range of Windows-specific inputs, including Windows Event Log, Windows Registry, WMI, Active Directory, and Performance monitoring.
Other sources	Other input sources are supported, such as FIFO queues and scripted inputs for getting data from APIs, and other remote data interfaces.

Well done! In this room, you learned about Splunk's core components, explored the Splunkinterface, and practiced uploading data to Splunk. You have gained the foundational knowledge of Splunk SIEM.

If you'd like to dig deeper, you can explore the following Splunk walkthrough and challenge rooms to understand how Splunk is effectively used in investigating incidents.

Splunk: Exploring SPL
Incident Handling with Splunk
Investigating With Splunk
Benign - Challenge
PoshEclipse - Challenge

Splunk is a powerful SIEM solution that provides the ability to search and explore machine data. **Search Processing Language (SPL)** is used to make the search more effective. It comprises various functions and commands used together to form complex yet effective search queries to get optimized results.

This room will dive deep into some key fundamentals of searching capability, like chaining SPL queries to construct simple to complex queries.

Learning Objectives

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What are Search processing Language?
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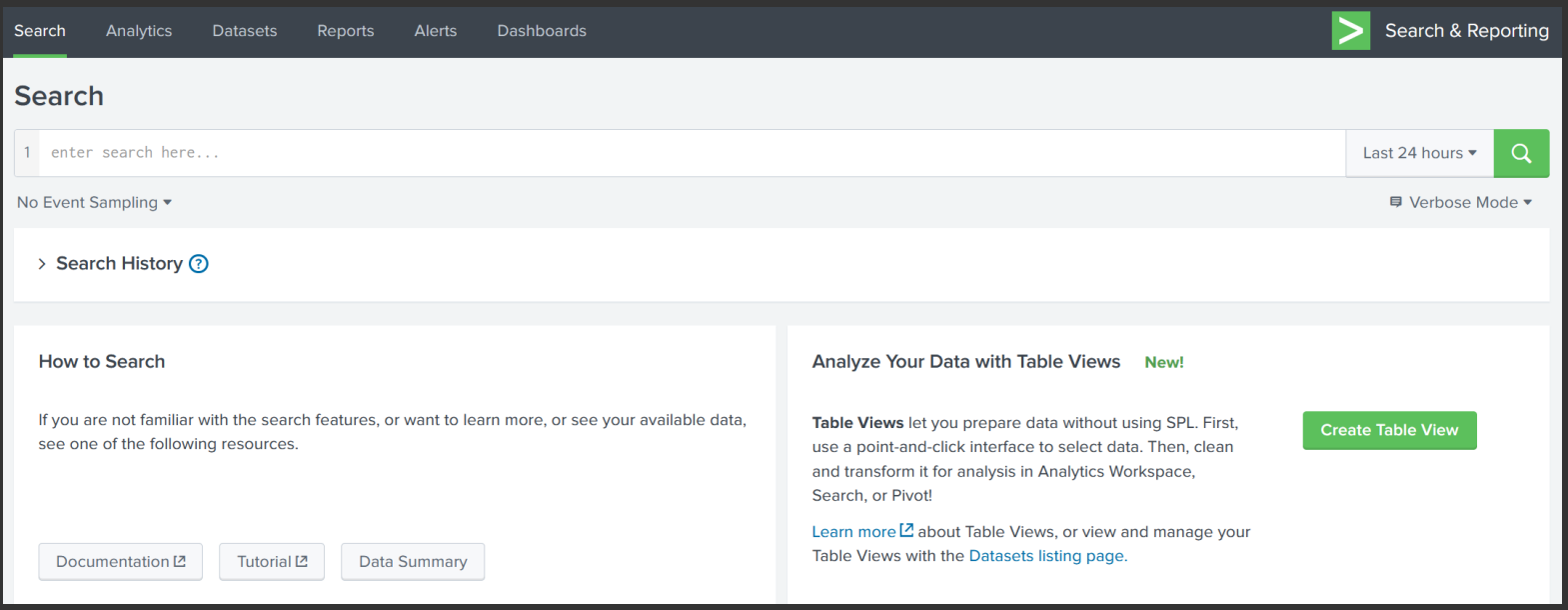
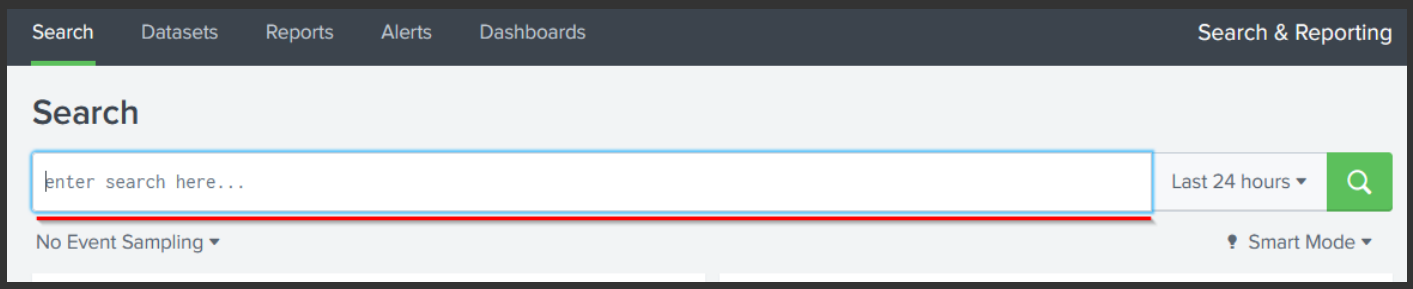
Search & Reporting App is the default interface used to search and analyze the data on the Splunk Home page. It has various functionalities that assist analysts in improving the search experience.



Some important functionalities present in the search App are explained below:

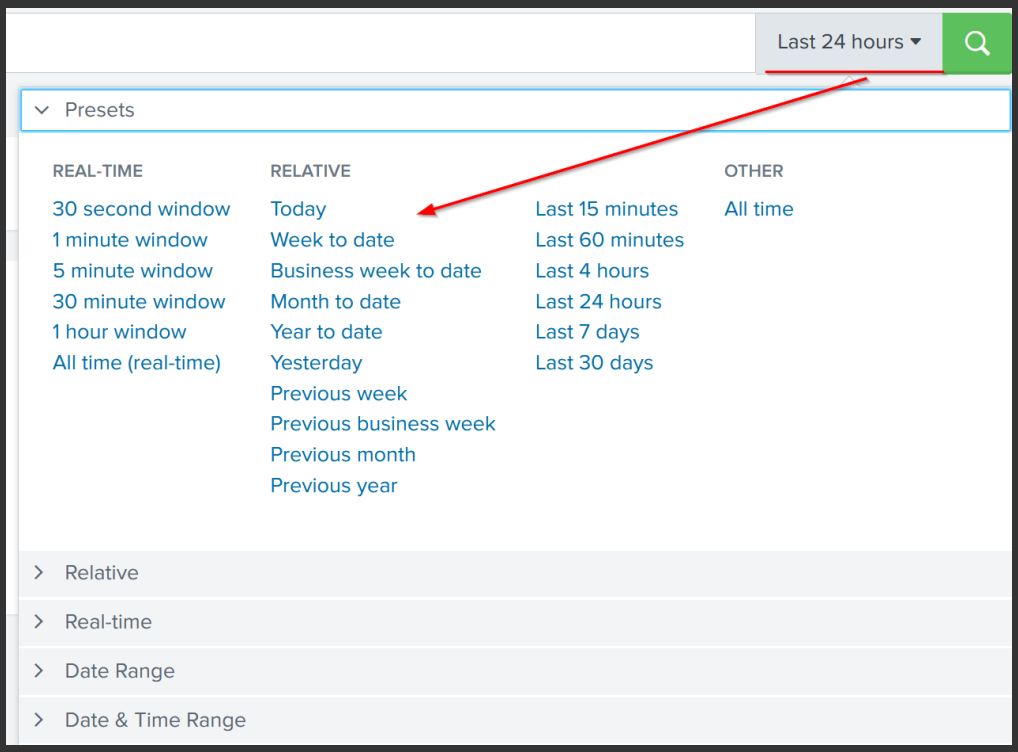
1) Search Head:

Search Head is where we use search processing language queries to look for the data



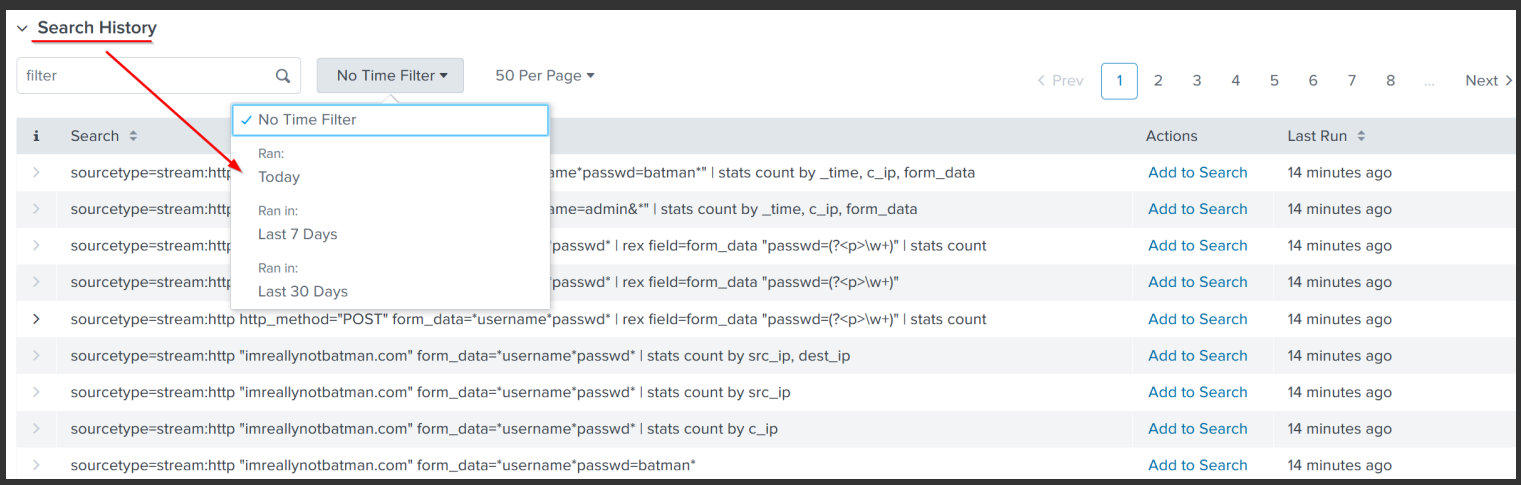
2) Time Duration:

This tab option provides multiple options to select the time duration for the search. **All-time** will display the events in real-time. Similarly, the **last 60 minutes** will display all the events captured in the last hour



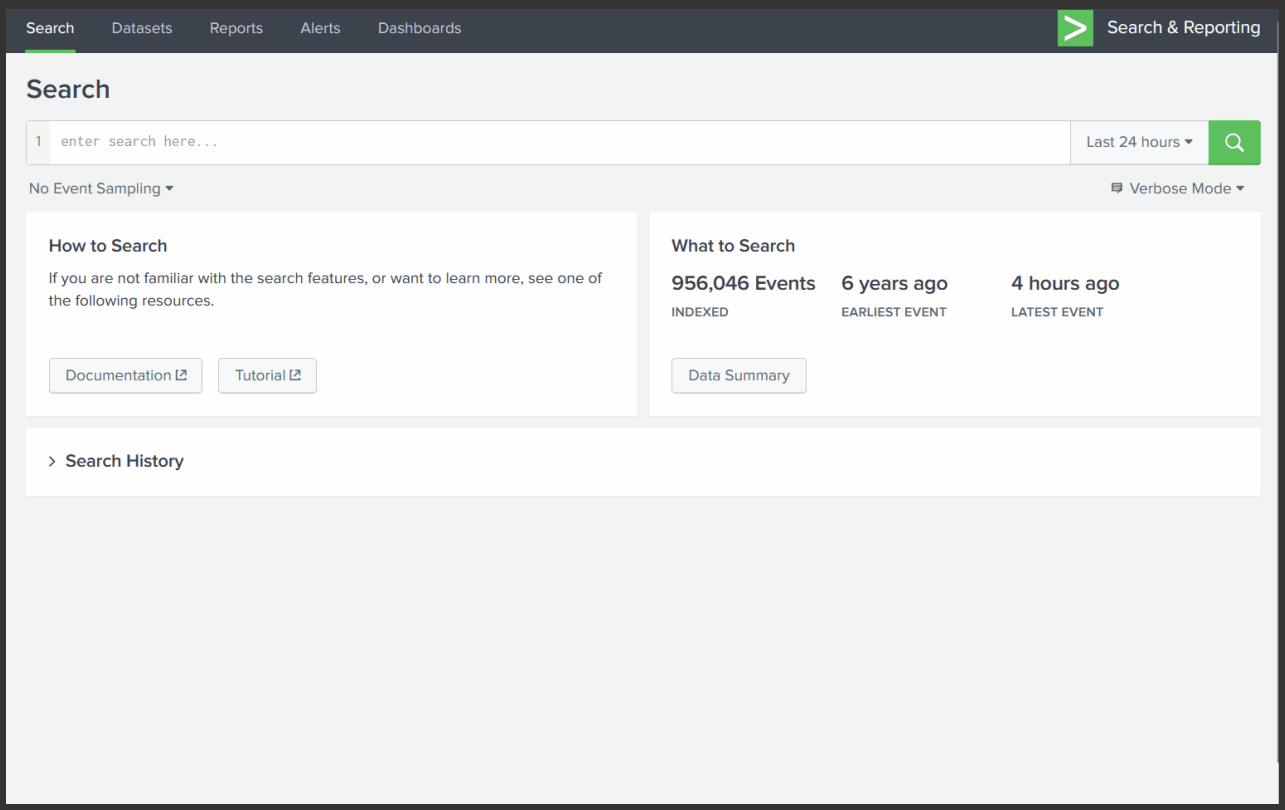
3) Search History:

This tab saves the search queries that the user has run in the past along with the time when it was run. It lets the user click on the past searches and look at the result. The filter option is used to search for the particular query based on the term.



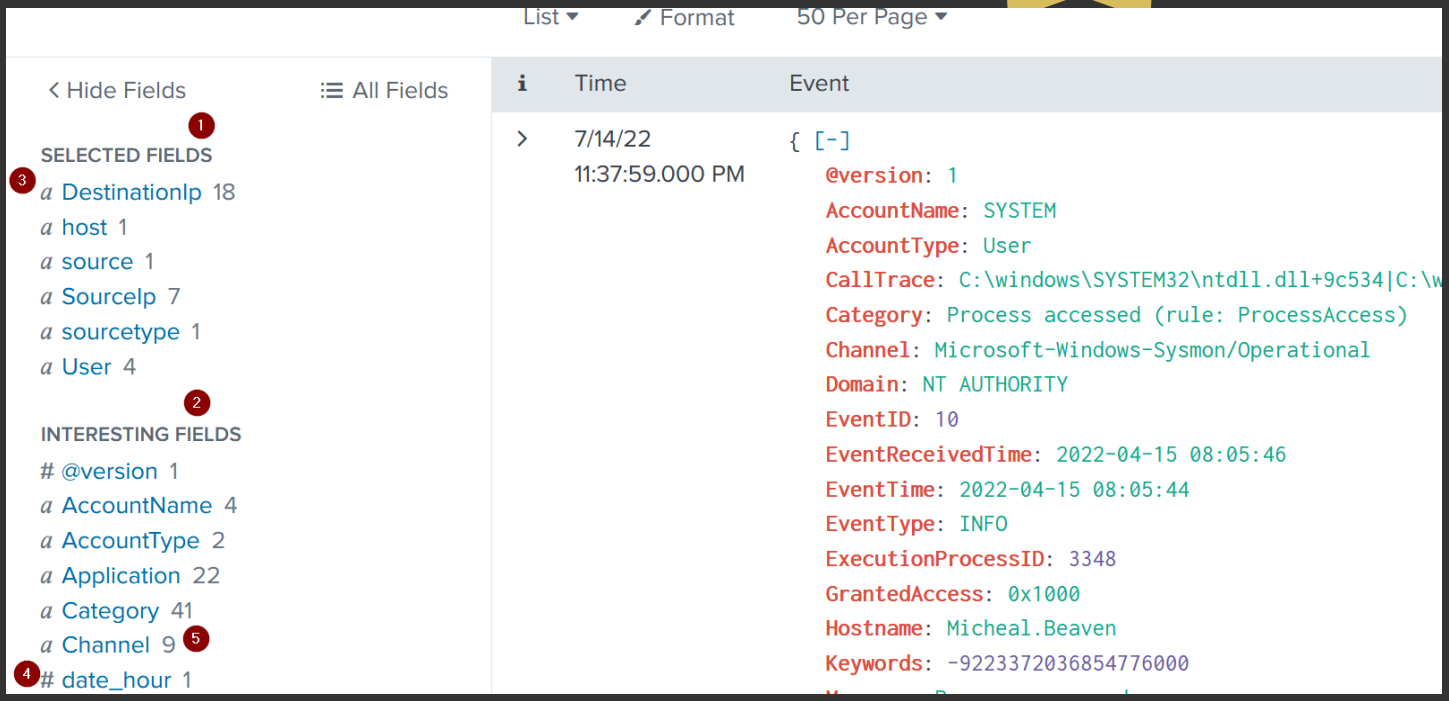
4) Data Summary:

This tab provides a summary of the data type, the data source, and the hosts that generated the events as shown below. This tab is very important feature used to get a brief idea about the network visibility.



5) Field Sidebar:

The Field Sidebar can be found on the left panel of Splunk search. This sidebar has two sections showing selected fields and interesting fields. It also provides quick results, such as top values and raw values against each field.



Some important points to understand about the sidebar are explained below:

1- Selected Fields	Splunk extracts the default fields like source, sourcetype, and host, which appear in each event, and places them under the selected fields column. We can select other fields that seem essential and add them to the list.
2- Interesting Fields	Pulls all the interesting fields it finds and displays them in the left panel to further explore.
3- Alpha-numeric fields 'a'	This alpha symbol shows that the field contains text values.
4- Numeric fields '#'	This symbol shows that this field contains numerical values.
5- Count	The number against each field shows the number of events captured in that timeframe.

Splunk Search Processing Language comprises of multiple functions, operators and commands that are used together to form a simple to complex search and get the desired results from the ingested logs. Main components of SPL are explained below:

Search Field Operators

Splunk field operators are the building blocks used to construct any search query. These field operators are used to filter, remove, and narrow down the search result based on the given criteria. Common field operators are Comparison operators, wildcards, and boolean operators.

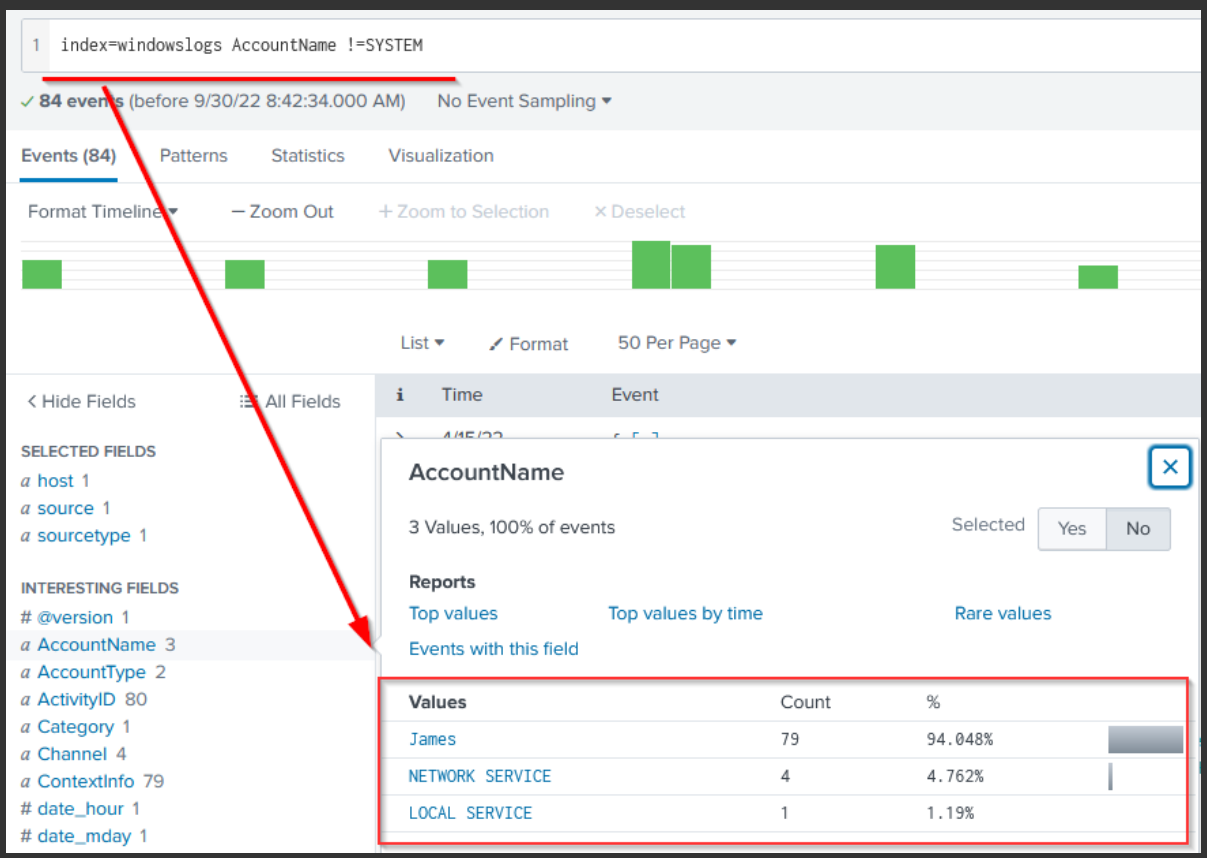
Comparison Operators

These operators are used to compare the values against the fields. Some common comparisons operators are mentioned below:

Field Name	Operator	Example	Explanation
Equal	=	UserName=Mark	This operator is used to match values against the field. In this example, it will look for all the events, where the value of the field UserName is equal to Mark.
Not Equal to	!=	UserName! =Mark	This operator returns all the events where the UserName value does not match Mark.
Less than	<	Age < 10	Showing all the events with the value of Age less than 10.
Less than or Equal to	<=	Age <= 10	Showing all the events with the value of Age less than or equal to 10.
Greater than	>	Outbound_traffic > 50 MB	This will return all the events where the Outbound traffic value is over 50 MB.
Greater Than or Equal to	>=	Outbound_traffic >= 50 MB	This will return all the events where the Outbound traffic value is greater or equal to 50 MB.

Lets use the comparison operator to display all the event logs from the index "windowslogs", where AccountName is not Equal to "System"

Search Query: `index=windowslogs AccountName !=SYSTEM`



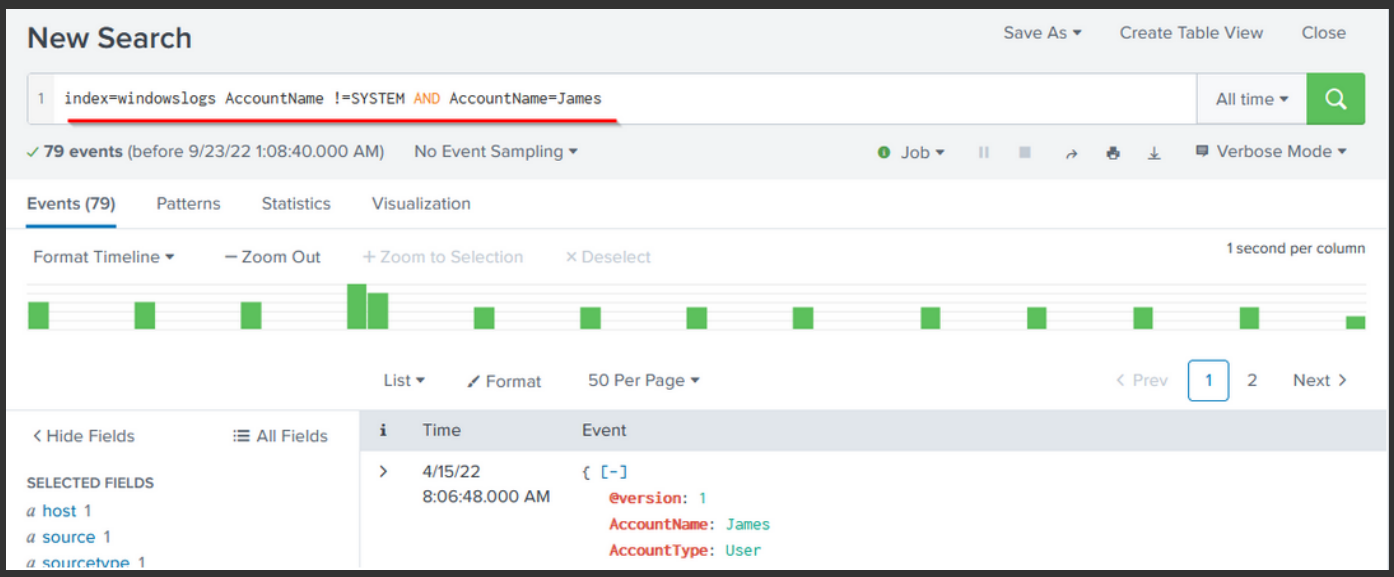
Boolean Operators

Splunk supports the following Boolean operators, which can be very handy in searching/filtering and narrowing down results.

Operator	Syntax	Explanation
NOT	field_A **NOT** value	Ignore the events from the result where field_A contain the specified value.
OR	field_A=value1 **OR** field_A=value2	Return all the events in which field_A contains either value1 or value2.
AND	field_A=value1 **AND** field_B=value2	Return all the events in which field_A contains value1 and field_B contains value2.

To understand how boolean operator works in SPL, lets add the condition to show the events from the James account.

Search Query: `index=windowslogs AccountName !=SYSTEM AND AccountName=James`



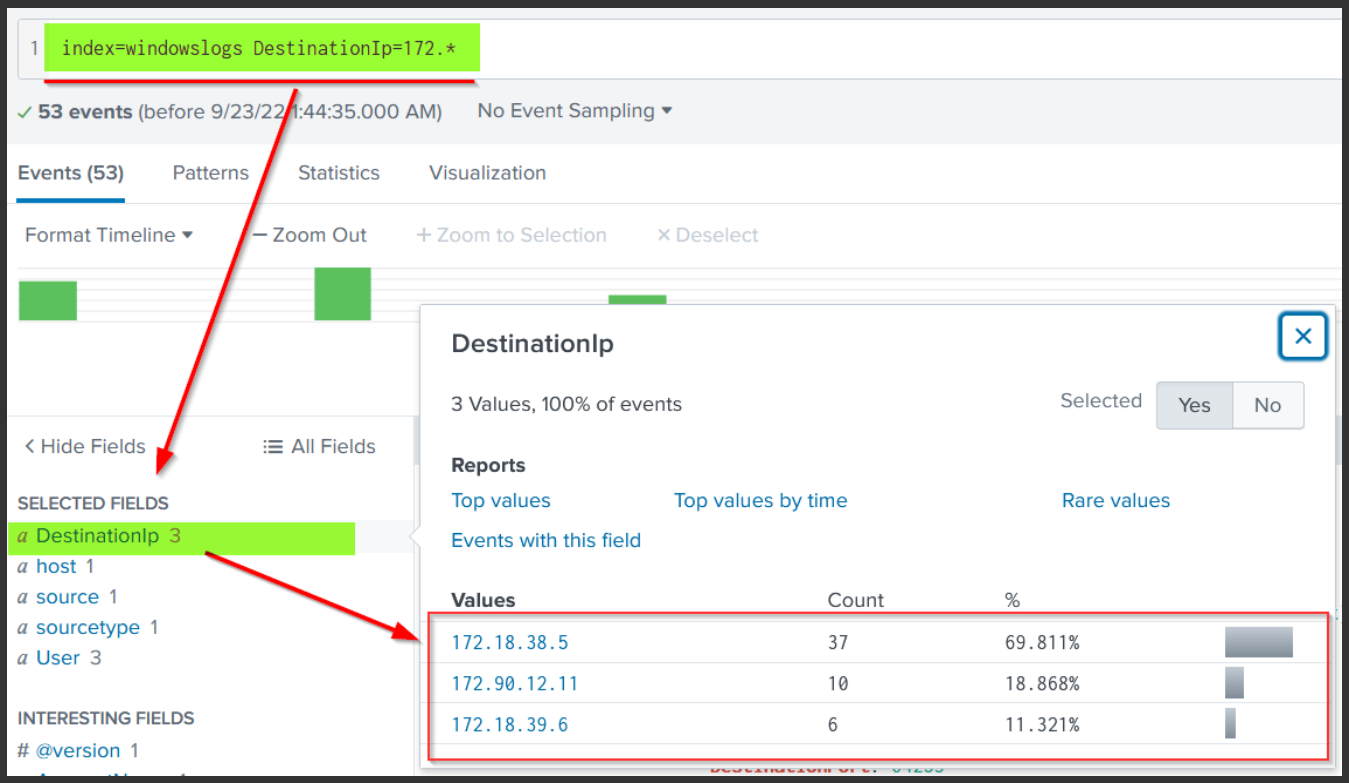
Wild Card

Splunk supports wildcards to match the characters in the strings.

Wildcard symbol	Example	Explanation
*	status=fail*	It will return all the results with values like status=failed status=failure

In the events, there are multiple DestinationIPs reported. Let's use the wildcard only to show the **DestinationIP** starting from 172.*

Search Query: `index=windowslogs DestinationIp=172.*`



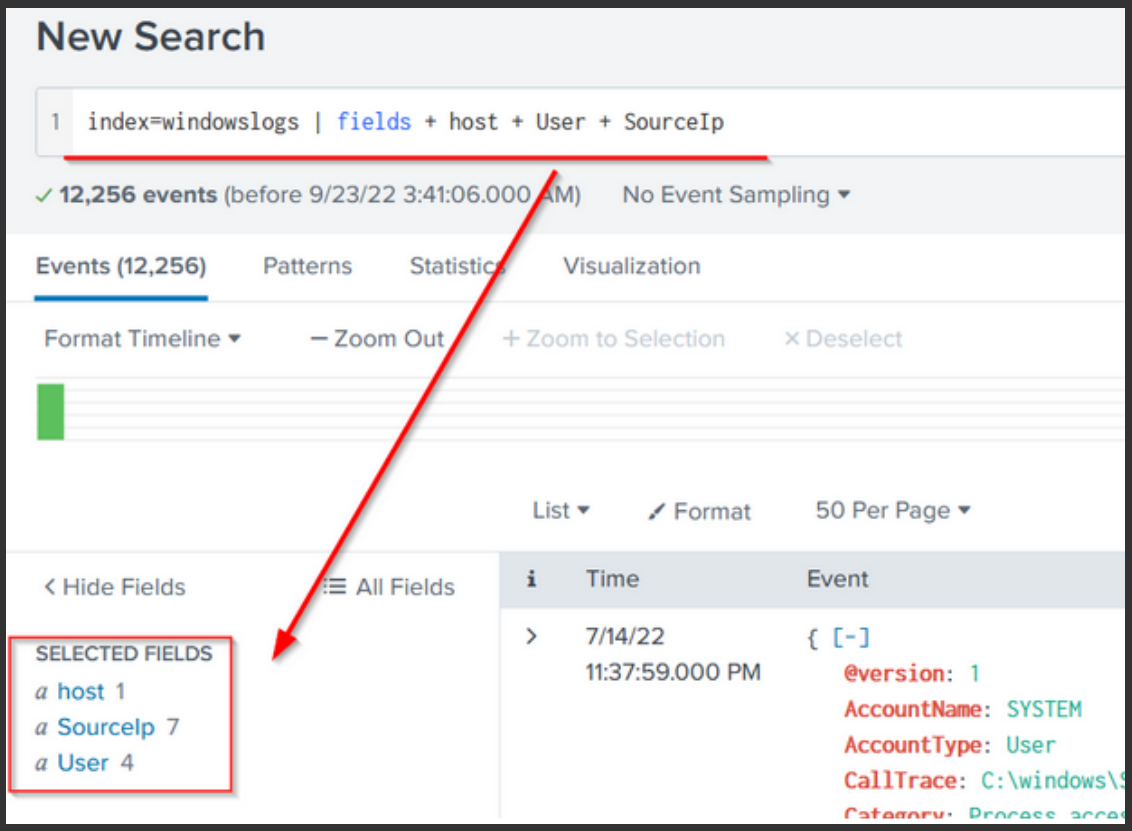
Our network generates thousands of logs each minute, all ingesting into our SIEM solution. It becomes a daunting task to search for any anomaly without using filters. SPL allows us to use **Filters** to narrow down the result and only show the important events that we are interested in. We can add or remove certain data from the result using filters. The following commands are useful in applying filters to the search results.

Fields

Command	**fields**
Explanation	Fields command is used to add or remove mentioned fields from the search results. To remove the field, minus sign (-) is used before the fieldname and plus (+) is used before the fields which we want to display.
Syntax	| fields <field_name1> <field_name2>
Example	`| fields + HostName - EventID`

Let's use the `fields` command to only display host, User, and SourceIP fields using the following syntax.

Search Query: `index=windowslogs | fields + host + User + SourceIp`



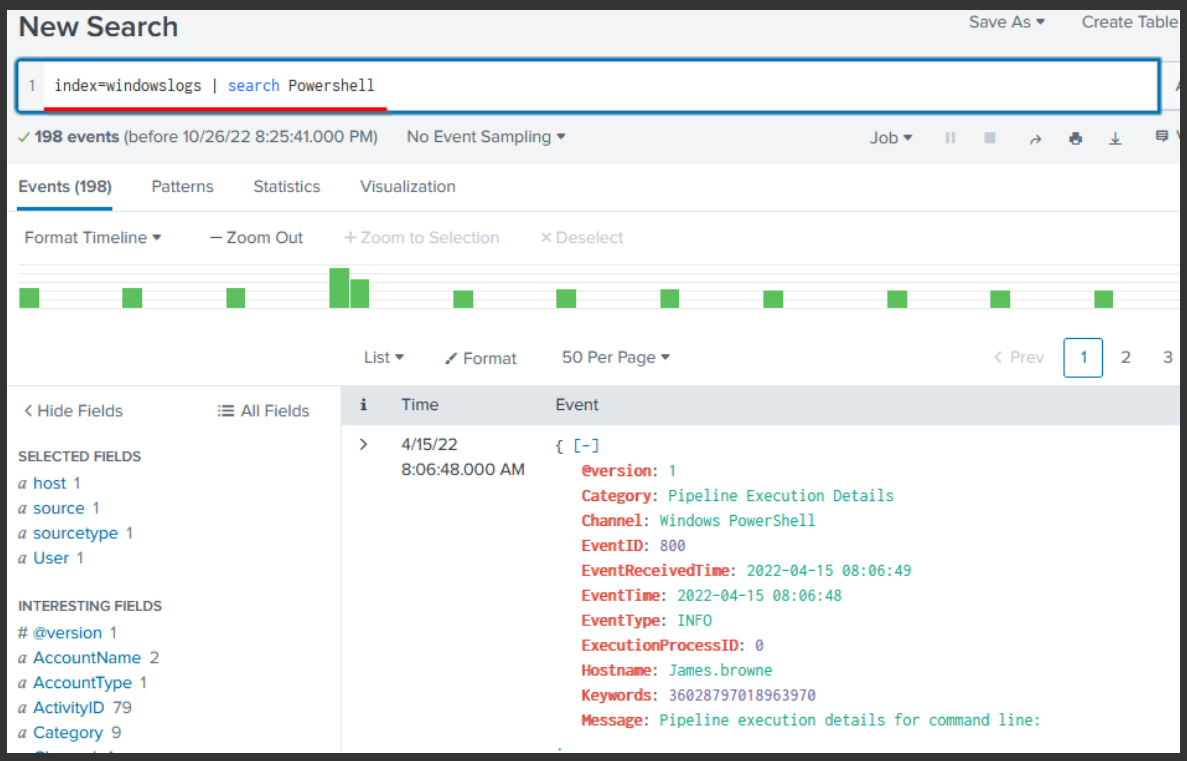
Note: Click on the **More field** to display the fields if some fields are not visible.

Search

Command	**search**
Explanation	This command is used to search for the raw text while using the chaining command |
Syntax	| search <search_keyword>
Example	`| search "Powershell"`

Use the `search` command to show all the events containing the term Powershell. This will return all the events that contain the term "**Powershell**".

Search Query: `index=windowslogs | search Powershell`



Dedup

Command	**dedup**
Explanation	Dedup is the command used to remove duplicate fields from the search results. We often get the results with various fields getting the same results. These commands remove the duplicates to show the unique values.
Syntax	| dedup <fieldname>
Example	`| dedup EventID`

We can use the `dedup` command to show the list of unique **EventIDs** from a particular hostname.

Search Query: `index=windowslogs | table EventID User Image Hostname | dedup EventID`

New Search

Save As | Create Table View | Close

```
index=windowslogs | table EventID User Image Hostname | dedup EventID
```

All time

✓ **12,256 events** (before 10/26/22 8:38:50.000 PM) No Event Sampling

Job | Verbose Mode

Events (12,256) | Patterns | **Statistics (55)** | Visualization

100 Per Page | Format | Preview

EventID	User	Image	Hostname
18		C:\Windows\System32\RuntimeBroker.exe	James.browne
12		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	James.browne
3	NT AUTHORITY\SYSTEM	C:\Windows\System32\dns.exe	Salena.Adam
22		C:\Windows\System32\svchost.exe	James.browne
7		C:\Windows\System32\svchost.exe	James.browne
1	NT AUTHORITY\SYSTEM	C:\Windows\System32\svchost.exe	James.browne

Rename

Command	**rename**
Explanation	It allows us to change the name of the field in the search results. It is useful in a scenario when the field name is generic or log, or it needs to be updated in the output.
Syntax	| rename <fieldname>
Example	`| rename User as Employees`

Let's rename the User field to Employees using the following search query.

Search Query: `index=windowslogs | fields + host + User + SourceIp | rename User as Employees`

New Search

```
index=windowslogs | fields + host + User + SourceIp | rename User as Employees
```

✓ **12,256 events** (before 9/23/22 3:45:00.000 AM) No Event Sampling

Events (12,256) | Patterns | Statistics | Visualization

Format Timeline | − Zoom Out | + Zoom to Selection | × Deselect

List | Format | 50 Per Page

< Hide Fields | All Fields

SELECTED FIELDS

a Employees 4

a host 1

a SourceIp 7

+ Extract New Fields

i	Time	Event
>	7/14/22 11:37:59.000 PM	{ [-] @version: 1 AccountName: SYSTEM AccountType: User CallTrace: C:\windows Category: Process acc Channel: Microsoft-Wi

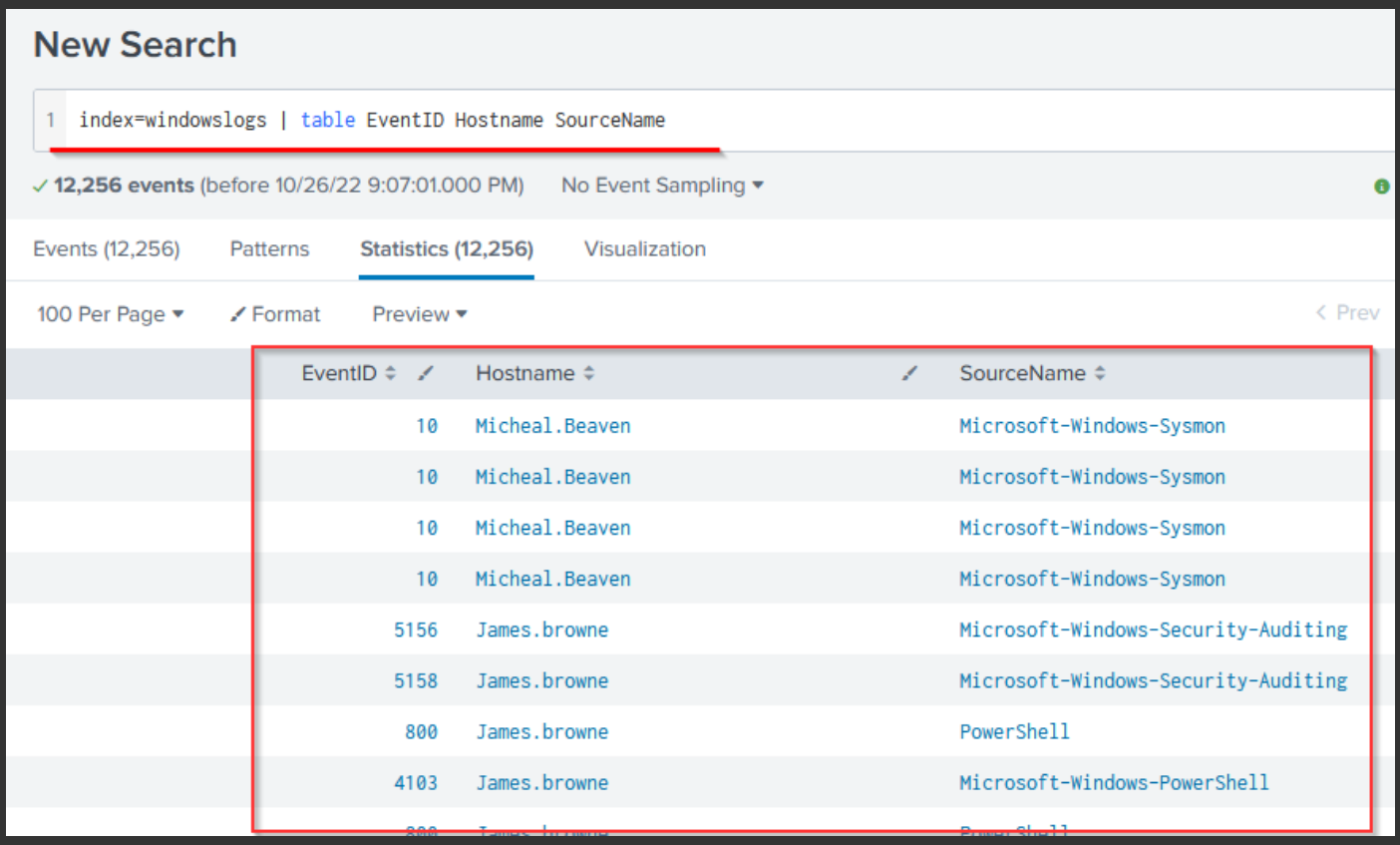
SPL provides various commands to bring structure or order to the search results. These sorting commands like `head`, `tail`, and `sort` can be very useful during logs investigation. These ordering commands are explained below:

Table

Explanation	Each event has multiple fields, and not every field is important to display. The Table command allows us to create a table with selective fields as columns.
Syntax	I table <field_name1> <fieldname_2>
Example	I `table` I `head 20` # will return the top 20 events from the result list.

This search query will create a table with three columns selected and ignore all the remaining columns from the display.

Search Query: `index=windowslogs | table EventID Hostname SourceName`

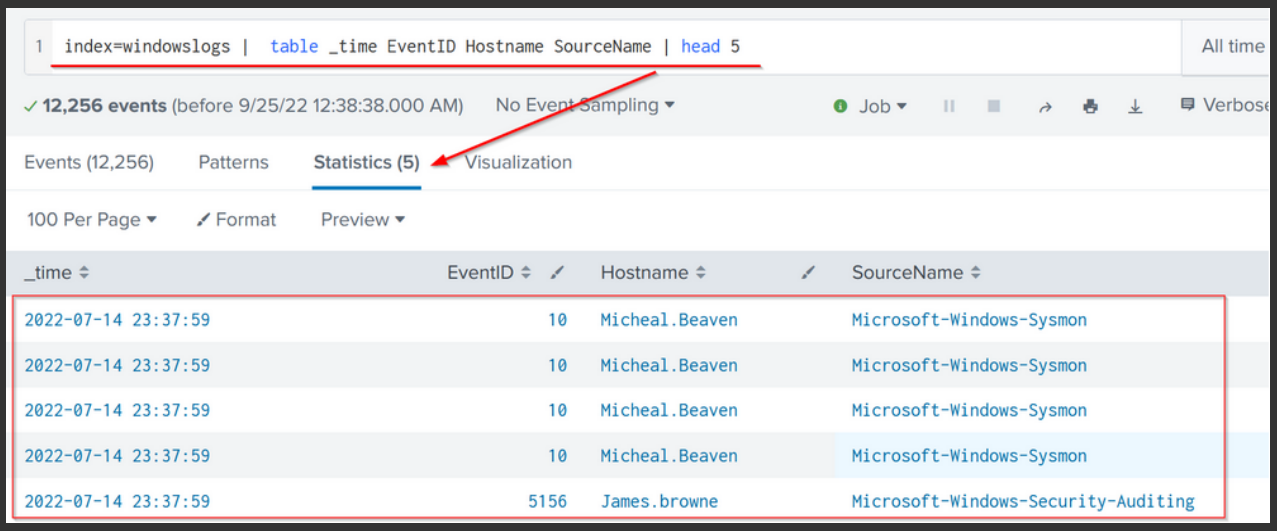


Head

Explanation	The **head** command returns the first 10 events if no number is specified.
Syntax	I head <number>
Example	I `head` # will return the top 10 events from the result list I `head 20` # will return the top 20 events from the result list

The following search query will show the table containing the mentioned fields and display only the top 5 entries.

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | head 5`



Tail

Explanation	The **Tail** command returns the last 10 events if no number is specified.
Syntax	I tail <number>
Example	I `tail` # will return the last 10 events from the result list I `tail 20` # will return the last 20 events from the result list

The following search query will show the table containing the mentioned fields and display only 5 entries from the bottom of the list.

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | tail 5`

```
index=windowslogs |  table _time EventID Hostname SourceName | tail 5
```

✓ **12,256 events** (before 9/25/22 1:14:18.000 AM) No Event Sampling ▾ Job ▾

Events (12,256) Patterns **Statistics (5)** Visualization

100 Per Page ▾ Format Preview ▾

_time	EventID	Hostname	SourceName
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800		PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell

sort

Explanation	The **Sort** command allows us to order the fields in ascending or descending order.
Syntax	`| sort <field_name>`
Example	`| sort Hostname` # This will sort the result in Ascending order.

The following search query will sort the results based on the Hostname field.

Search Query: `index=windowslogs |  table _time EventID Hostname SourceName | sort Hostname`

```
index=windowslogs |  table _time EventID Hostname SourceName | sort Hostname
```

✓ **12,256 events** (before 9/25/22 1:17:18.000 AM) No Event Sampling ▾ Job ▾

Events (12,256) Patterns **Statistics (10,000)** Visualization

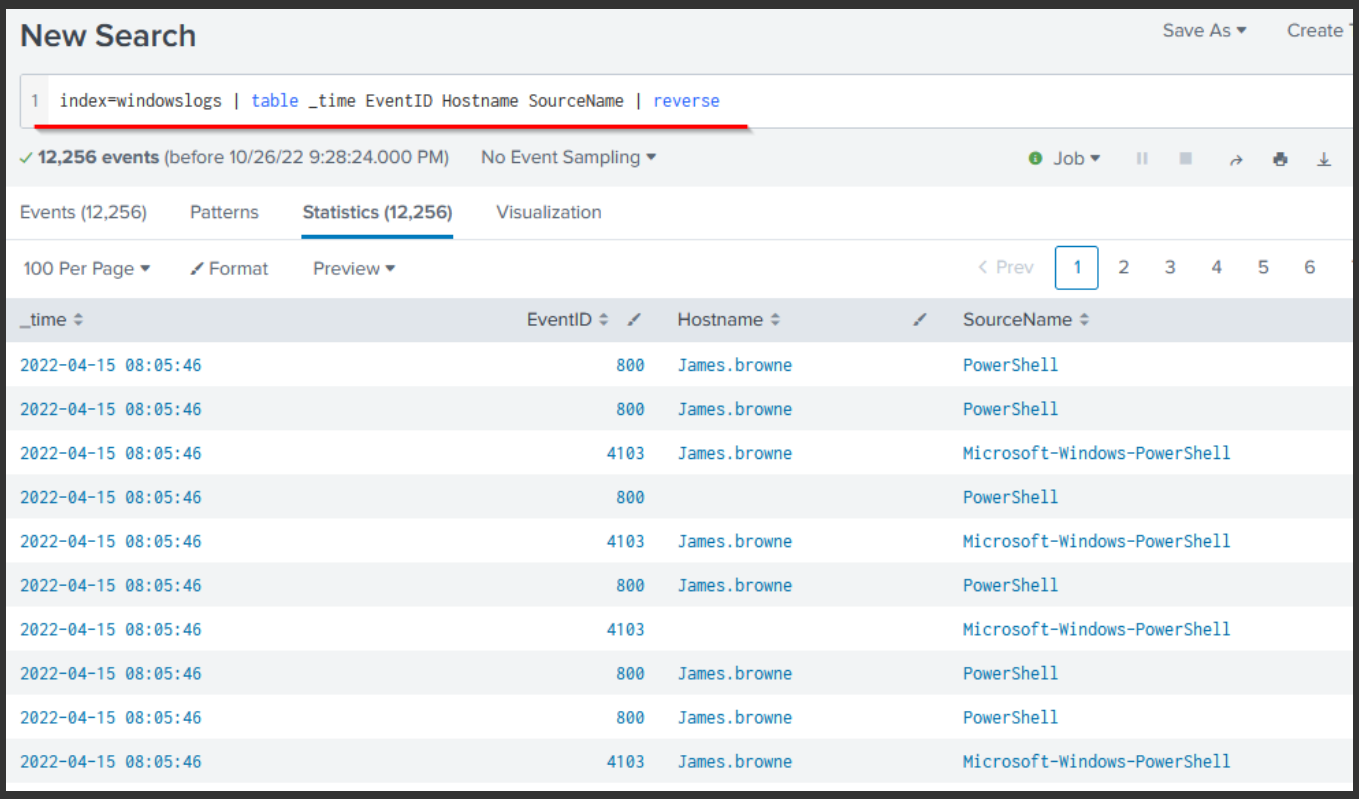
100 Per Page ▾ Format Preview ▾ < Prev 1 2 3 4 5 6 7 8

_time	EventID	Hostname	SourceName
2022-07-14 23:37:59	5156	James.browne	Microsoft-Windows-Security-Auditing
2022-07-14 23:37:59	5158	James.browne	Microsoft-Windows-Security-Auditing
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell

Reverse

Explanation	The reverse command simply reverses the order of the events.
Syntax	| reverse
Example	`<Search Query> | reverse`

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | reverse`



_time	EventID	Hostname	SourceName
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800		PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103		Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell

Transformational commands are those commands that change the result into a data structure from the field-value pairs. These commands simply transform specific values for each event into numerical values which can easily be utilized for statistical purposes or turn the results into visualizations. Searches that use these transforming commands are called transforming searches. Some of the most used transforming commands are explained below.

General Transformational Commands

Top

Command	**top**
Explanation	This command returns frequent values for the top 10 events.
Syntax	I top <field_name> I top limit=6 <field_name>
Example	`top limit=3 EventID`

The following command will display the top 7 Image (representing Processes) captured.

Search Query: `index=windowslogs | top limit=7 Image`

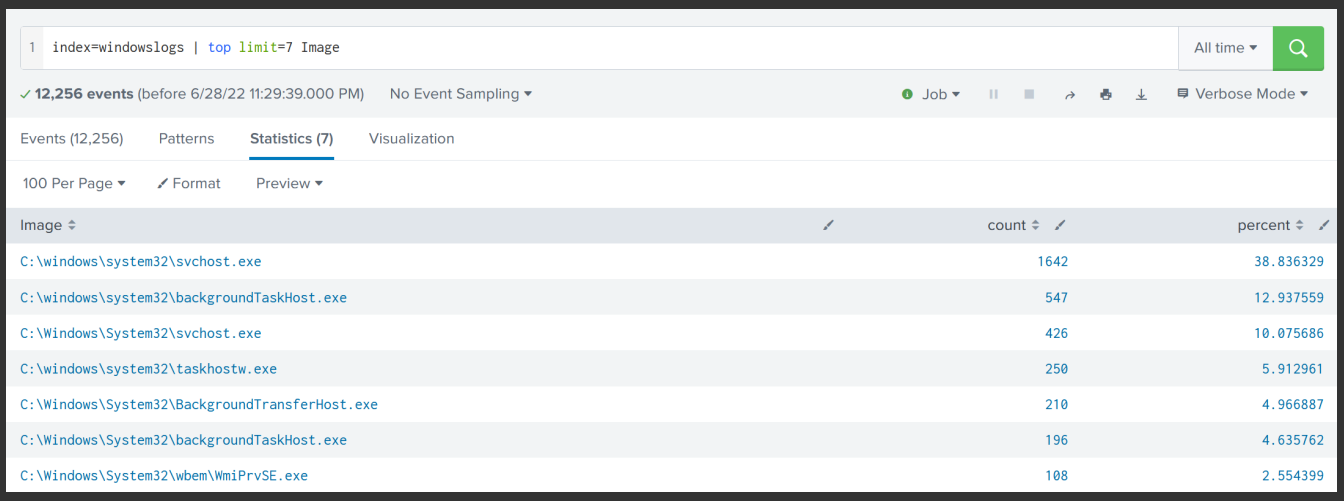


Image	count	percent
C:\windows\system32\svchost.exe	1642	38.836329
C:\windows\system32\backgroundTaskHost.exe	547	12.937559
C:\Windows\System32\svchost.exe	426	10.075686
C:\windows\system32\taskhostw.exe	250	5.912961
C:\Windows\System32\BackgroundTransferHost.exe	210	4.966887
C:\Windows\System32\backgroundTaskHost.exe	196	4.635762
C:\Windows\System32\wbem\WmiPrvSE.exe	108	2.554399

Rare

Command	**rare**
Explanation	This command does the opposite of top command as it returns the least frequent values or bottom 10 results.
Syntax	I rare <field_name> I rare limit=6 <field_name>
Example	`rare limit=3 EventID`

The following command will display the rare 7 Image (Processes) captured.

Search Query: `index=windowslogs | rare limit=7 Image`

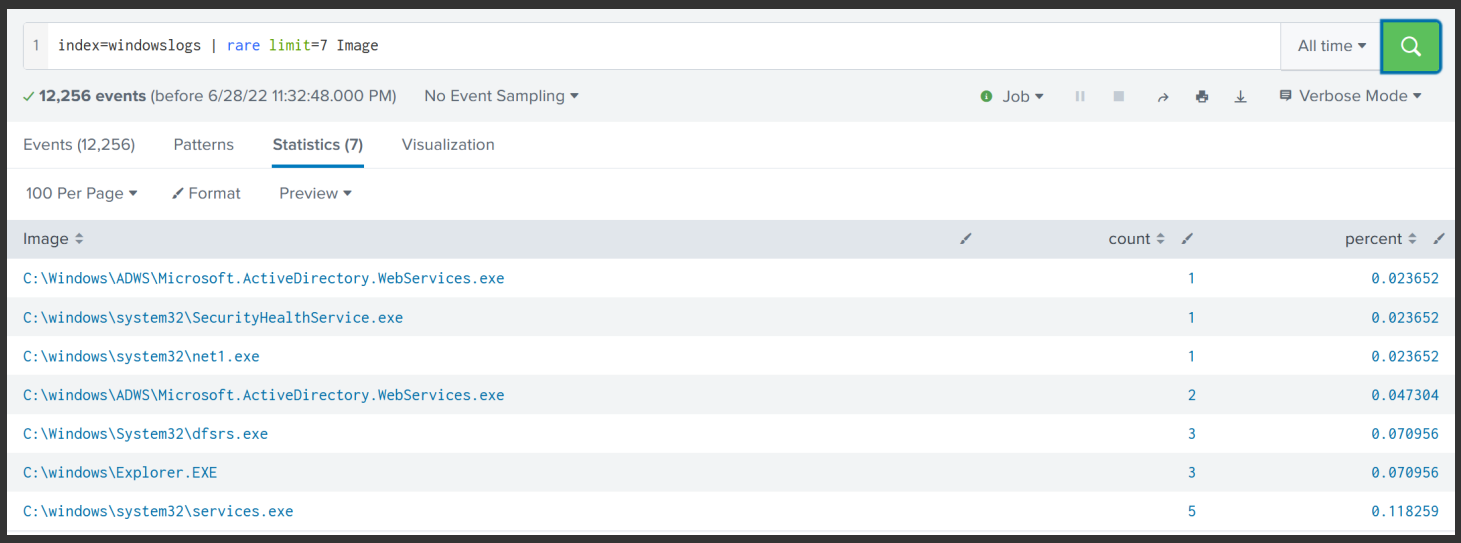


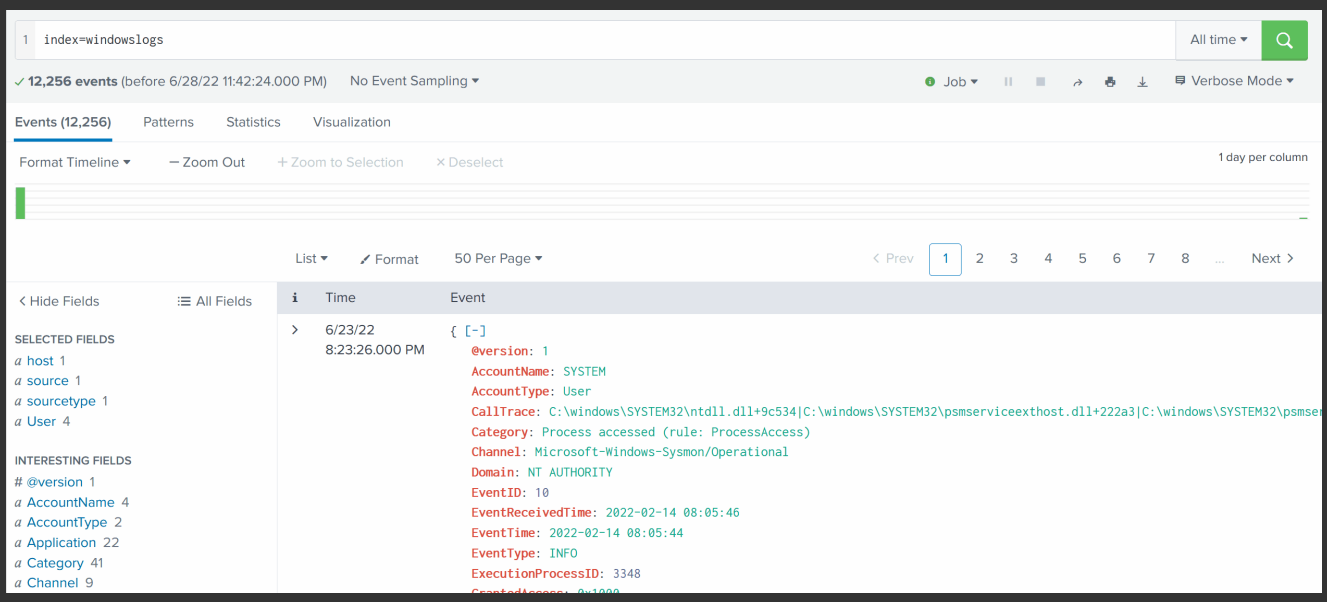
Image	count	percent
C:\Windows\ADWS\Microsoft.ActiveDirectory.WebServices.exe	1	0.023652
C:\windows\system32\SecurityHealthService.exe	1	0.023652
C:\windows\system32\net1.exe	1	0.023652
C:\windows\ADWS\Microsoft.ActiveDirectory.WebServices.exe	2	0.047304
C:\Windows\System32\dfsrs.exe	3	0.070956
C:\windows\Explorer.EXE	3	0.070956
C:\windows\system32\services.exe	5	0.118259

Highlight

Command	**highlight**
Explanation	The highlight command shows the results in raw events mode with fields highlighted.
Syntax	highlight <field_name1> <field_name2>
Example	`highlight User, host, EventID, Image`

The following command will highlight the three mentioned fields in the raw logs

Search Query: `index=windowslogs | highlight User, host, EventID, Image`



STATS Commands

SPL supports various stats commands that help in calculating statistics on the values. Some common stat commands are:

Command	Explanation	Syntax	Example
Average	This command is used to calculate the average of the given field.	stats avg(field_name)	stats avg(product_price)
Max	It will return the maximum value from the specific field.	stats max(field_name)	stats max(user_age)
Min	It will return the minimum value from the specific field.	stats min(field_name)	stats min(product_price)
Sum	It will return the sum of the fields in a specific value.	stats sum(field_name)	stats sum(product_cost)
Count	The count command returns the number of data occurrences.	stats count(function) AS new_NAME	stats count(source_IP)

Splunk Chart Commands

These are very important types of transforming commands that are used to present the data in table or visualization form. Most of the chart commands utilize various stat commands.

Chart

Command	**chart**
Explanation	The chart command is used to transform the data into tables or visualizations.
Syntax	| chart <function>
Example	| `chart count by User`

Search Query: `index=windowslogs | chart count by User`

Timechart

Command	**timechart**
Explanation	The timechart command returns the time series chart covering the field following the function mentioned. Often combined with STATS commands.
Syntax	| timechart function <field_name>
Example	| `timechart count by Image`

The following query will display the Image chart based on the time.

Search Query: `index=windowslogs | timechart count by Image`

This wraps up the Splunk SPL room.

We have covered various functions and commands used together to form search queries to look for the data more effectively. To practice more, look at the following Splunk rooms and challenges.

Incident handling with Splunk
Investigating with Splunk
Benign
PS Eclipse